

Preliminary Model Comparison Results

NuclearDetonation.jl: Lagrangian Particle Dispersion

NuclearDetonation.jl

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1 Model Overview

NuclearDetonation.jl is a Lagrangian particle dispersion model for simulating atmospheric transport of radioactive debris from nuclear detonations. Particles are released into a three-dimensional wind field (ERA5 reanalysis on the full 137-level hybrid sigma-pressure grid) and advected forward in time, with sub-grid turbulence, gravitational settling, and dry deposition acting on each particle individually.

Particles are distributed across a three-layer mushroom cloud structure (lower stem, mid-cap, upper cap) whose geometry and mass fractions are determined independently for each test by BIPOP-CMA-ES optimisation against digitised fallout observations. The test-specific release geometries are shown in Sections 2 and 5.

1.1 Ornstein–Uhlenbeck Turbulence

Atmospheric turbulence is represented by an Ornstein–Uhlenbeck (OU) stochastic process [2] rather than a simple random walk. In a random walk, each turbulent velocity perturbation is drawn independently at every timestep, producing unrealistically jagged trajectories. The OU process treats the turbulent velocity as a variable with temporal memory: it decays exponentially towards zero with a Lagrangian timescale τ_L , while a stochastic forcing term continually injects energy:

$$u'(t + \Delta t) = u'(t) \exp(-\Delta t / \tau_L) + \sigma_u \sqrt{1 - \exp(-2 \Delta t / \tau_L)} \xi, \quad (1)$$

where σ_u is the turbulent velocity standard deviation and $\xi \sim \mathcal{N}(0, 1)$. The standard deviations σ_u , σ_v , σ_w and timescales τ_L are computed from the Hanna (1982) parameterisation, which uses boundary-layer stability class and height above ground. Separate OU processes are maintained for the horizontal and vertical components.

The practical effect is that particles “remember” their turbulent velocity over timescales of tens of seconds to minutes, producing smoother, more physically plausible dispersion than an uncorrelated walk.

1.2 Gravitational Settling

Each particle is assigned a diameter drawn from a bimodal lognormal distribution (fine and coarse fractions), representing the range of debris sizes from sub-micron fission products to hundreds-of-micron soil particles. The gravitational settling velocity for each particle is computed from Stokes drag with a Cunningham slip correction [3] for small particles:

$$v_g = \frac{\rho_p d_p^2 g C_s}{18 \mu} \quad (2)$$

where ρ_p is the particle density, d_p is the diameter, μ is the dynamic viscosity of air (Sutherland's formula [4], temperature-dependent), and C_s is the Cunningham slip correction factor:

$$C_s = 1 + \frac{2\lambda}{d_p} \left(1.257 + 0.4 \exp\left(-\frac{0.55 d_p}{\lambda}\right) \right) \quad (3)$$

with λ the mean free path of air. The settling velocity is added to the vertical component of the particle velocity at each timestep, so large particles fall out quickly and deposit close to ground zero whilst fine particles remain aloft and travel long distances.

1.3 Dry Deposition

Surface removal of particles uses a resistance analogy following Zhang et al. [5]. The deposition velocity is:

$$v_d = v_g + \frac{1}{R_a + R_s} \quad (4)$$

where R_a is the aerodynamic resistance (from Monin–Obukhov similarity theory, accounting for atmospheric stability) and R_s is the surface resistance. The surface resistance depends on three collection mechanisms: Brownian diffusion ($E_B \propto Sc^{-0.54}$), inertial impaction (E_{IM} via the Stokes number), and interception (E_{IN} via the ratio of particle diameter to collector radius). A rebound correction $R_1 = 1 - \exp(-\sqrt{St})$ accounts for particles bouncing off surfaces rather than sticking.

When a particle's height drops below a threshold near the surface, its mass is reduced exponentially at a rate $k = v_d/h_{\text{mix}}$, transferring activity to the ground-level deposition field.

2 Nancy 24 kT: Consortium Comparison

Figure 1 shows the three-layer release geometry for the Nancy test, derived from NOAA 1984 debris cloud observations. The BIPOP-CMA-ES optimisation places 5.6% of activity in the lower stem (0–3.8 km), 35.1% in the mid-cap (3.8–6.1 km), and 59.3% in the upper cap (6.1–12.5 km). The Glasstone & Dolan (1977) empirical cloud dimensions for 24 kT provide the wireframe outline.

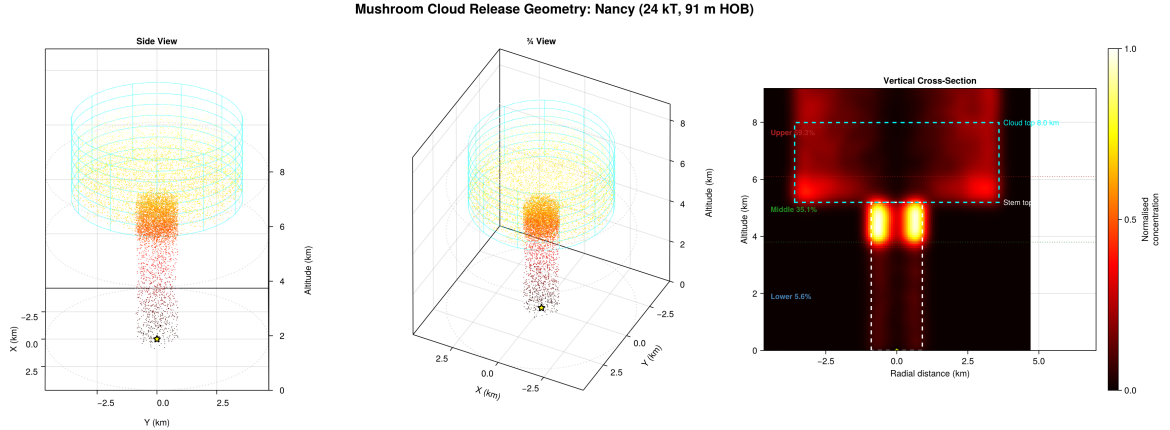


Figure 1: Nancy 24 kT release geometry. Three-layer particle distribution (lower 5.6%, middle 35.1%, upper 59.3%) with Glasstone & Dolan wireframe. Left/centre: 3D views; right: vertical cross-section with kernel density concentration field.

The model was first validated against the Upshot-Knothole Nancy test (24 kT, 24 March 1953). Simulated dose rate contours at H+12 hours are compared with digitised observations using the Figure of Merit in Space:

$$\text{FMS} = \frac{A_{\text{obs}} \cap A_{\text{model}}}{A_{\text{obs}} \cup A_{\text{model}}} \quad (5)$$

Six dose rate thresholds are evaluated: 0.4, 1, 4, 10, 40, and 100 mR/h. Each consortium member submitted output in their native format, interpolated onto a common 2 km grid for scoring.

Table 1 summarises the results. Our model achieves a mean FMS of 45.9%, roughly 17 percentage points above the next-best consortium member (UKMO, 28.8%).

Table 1: FMS scores for Nancy 24 kT at H+12.

Dose Rate (mR/h)	Our Model	UKMO	BfS	KIT	DSA/MET	DEMA
0.4	67.1%	60.4%	52.9%	33.8%	45.6%	0.7%
1	58.7%	47.3%	31.1%	27.7%	28.1%	—
4	55.6%	22.4%	18.2%	18.9%	16.9%	—
10	40.1%	15.9%	9.9%	12.1%	9.2%	—
40	28.7%	16.2%	5.4%	4.4%	5.5%	—
100	25.2%	10.5%	2.8%	0.5%	2.2%	—
Mean	45.9%	28.8%	20.0%	16.2%	17.9%	0.1%

3 Side-by-Side Comparisons

The following figures show the digitised observation contours (left panel) and each model's predicted dose rate contours (right panel) for the six thresholds.

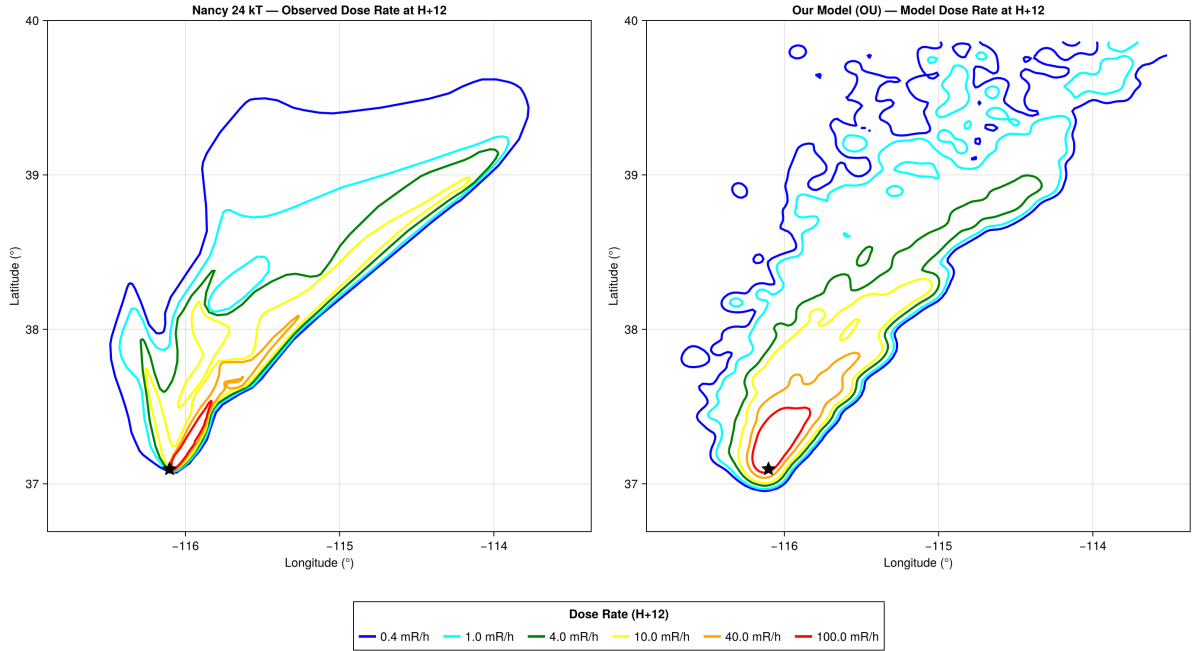


Figure 2: Nancy FMS comparison: our model (right) vs digitised observations (left) for six dose rate thresholds. Mean FMS 45.9%.

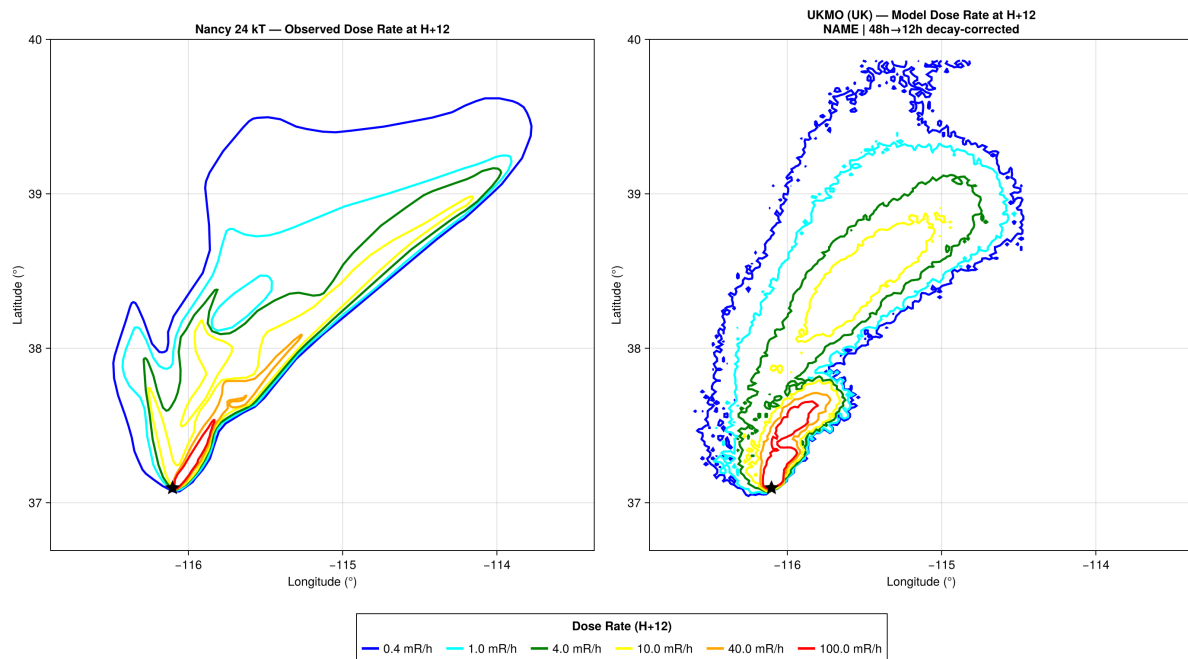


Figure 3: Nancy FMS comparison: UKMO, UK Met Office NAME model (right) vs digitised observations (left). Mean FMS 28.8%.

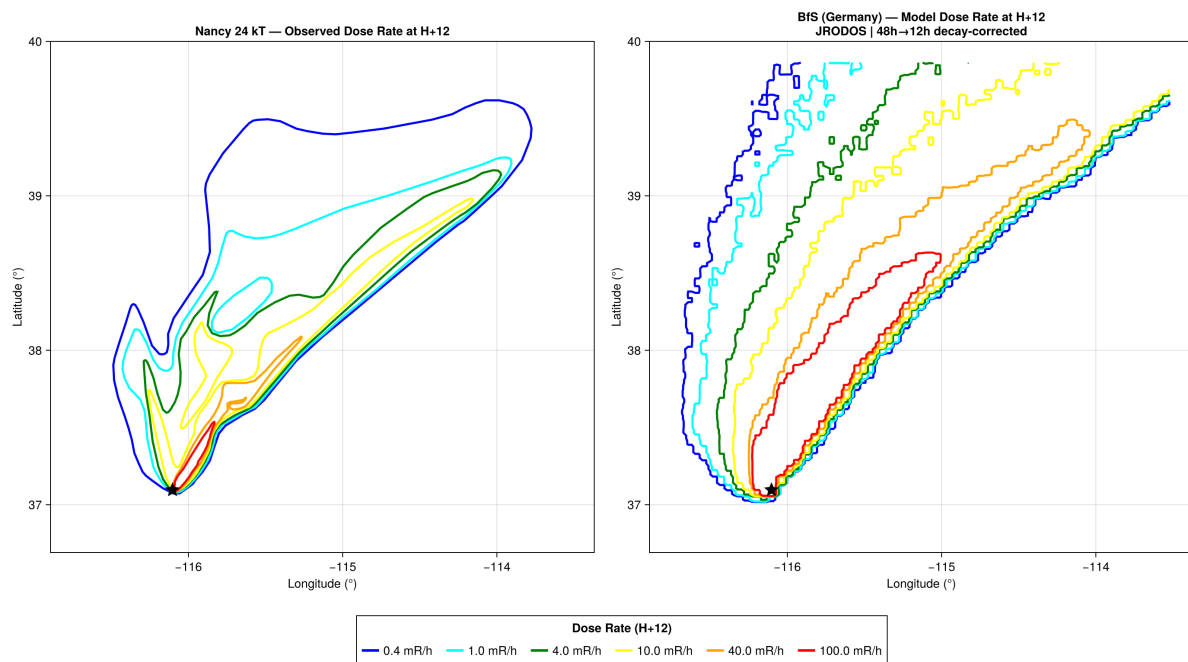


Figure 4: Nancy FMS comparison: BfS, Germany, JRODOS (right) vs digitised observations (left). Mean FMS 20.0%.

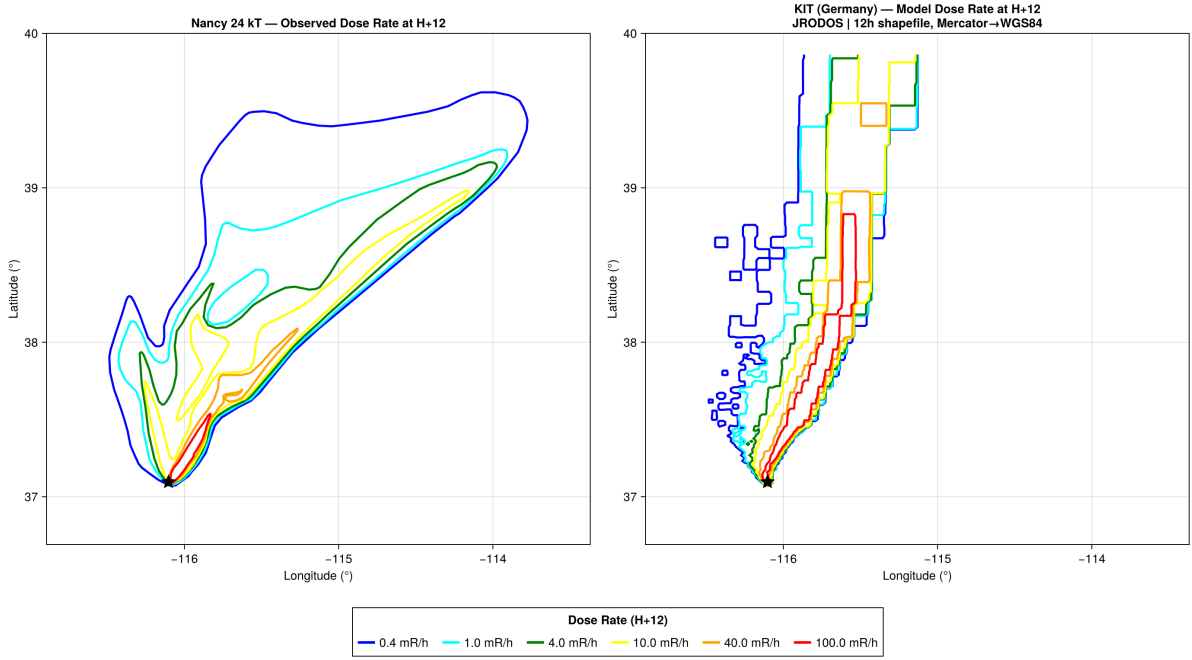


Figure 5: Nancy FMS comparison: KIT, Germany, JRODOS (right) vs digitised observations (left). Mean FMS 16.2%.

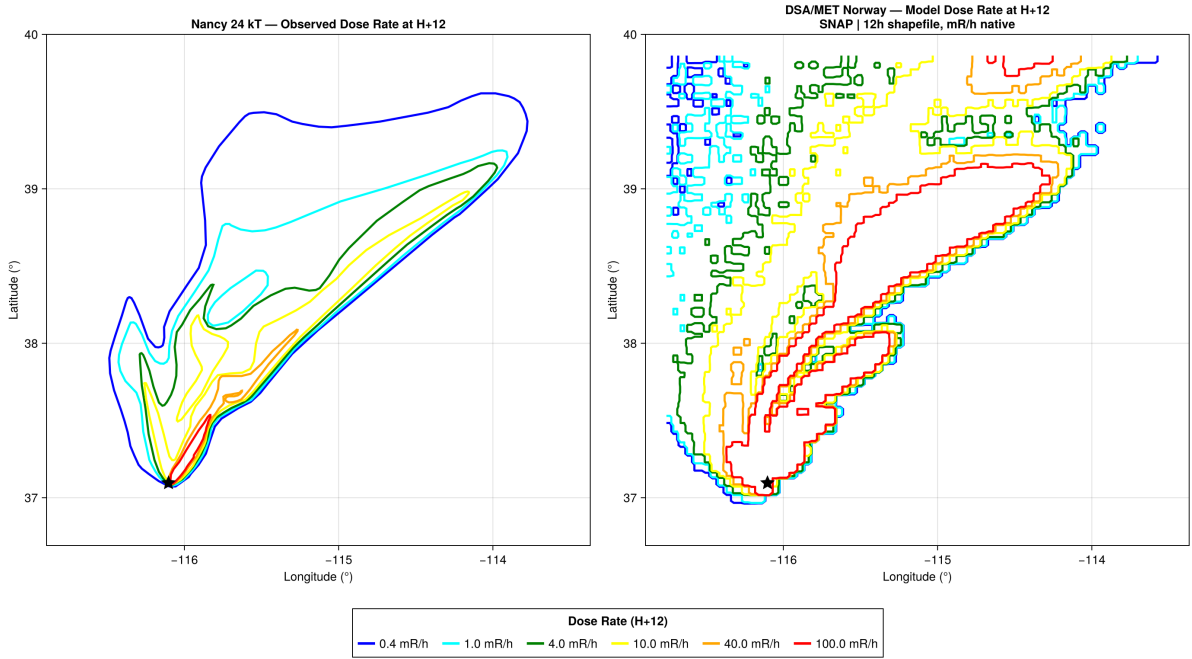


Figure 6: Nancy FMS comparison: DSA/MET Norway, SNAP model (right) vs digitised observations (left). Mean FMS 17.9%.

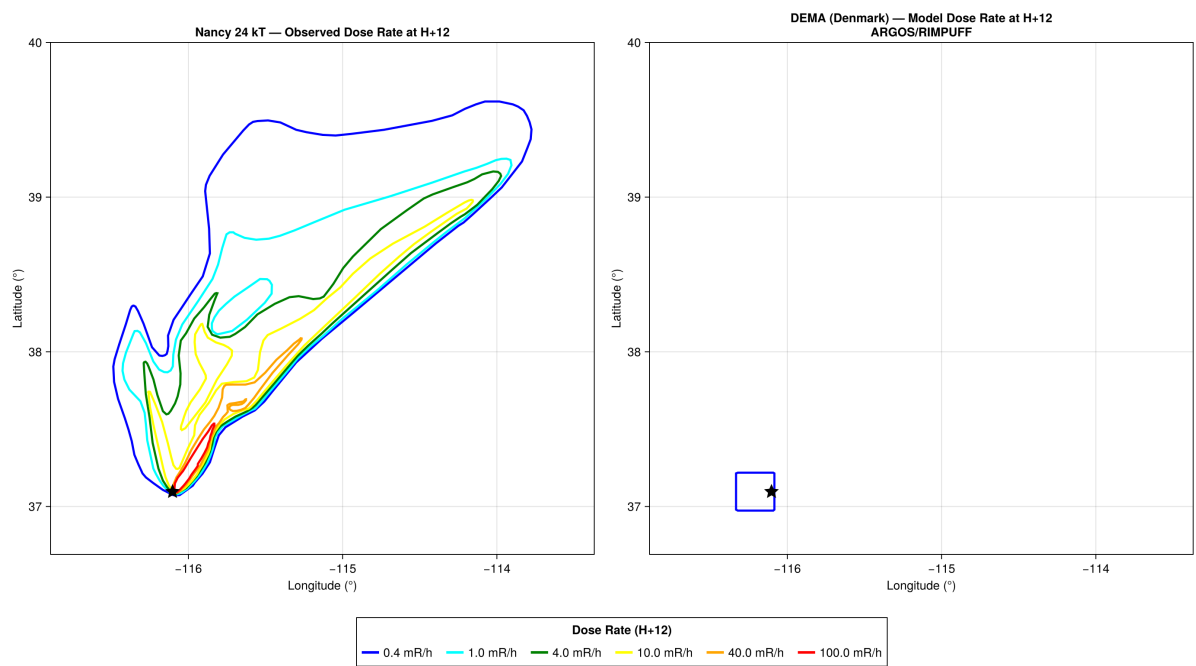


Figure 7: Nancy FMS comparison: DEMA, Denmark, ARGOS/RIMPUFF (right) vs digitised observations (left). Mean FMS 0.1%.

4 Nancy 24 kT: Observed vs Model

Figure 8 shows a direct comparison between digitised observations and model predictions for the Nancy test. The top row compares dose rate contours at H+12; the bottom row compares time-of-arrival (TOA) contours showing when the fallout plume first reached each location. The model TOA is computed from cumulative hourly deposition snapshots, smoothed with a Gaussian kernel and thresholded at 0.1% of the peak to detect the advancing plume front.

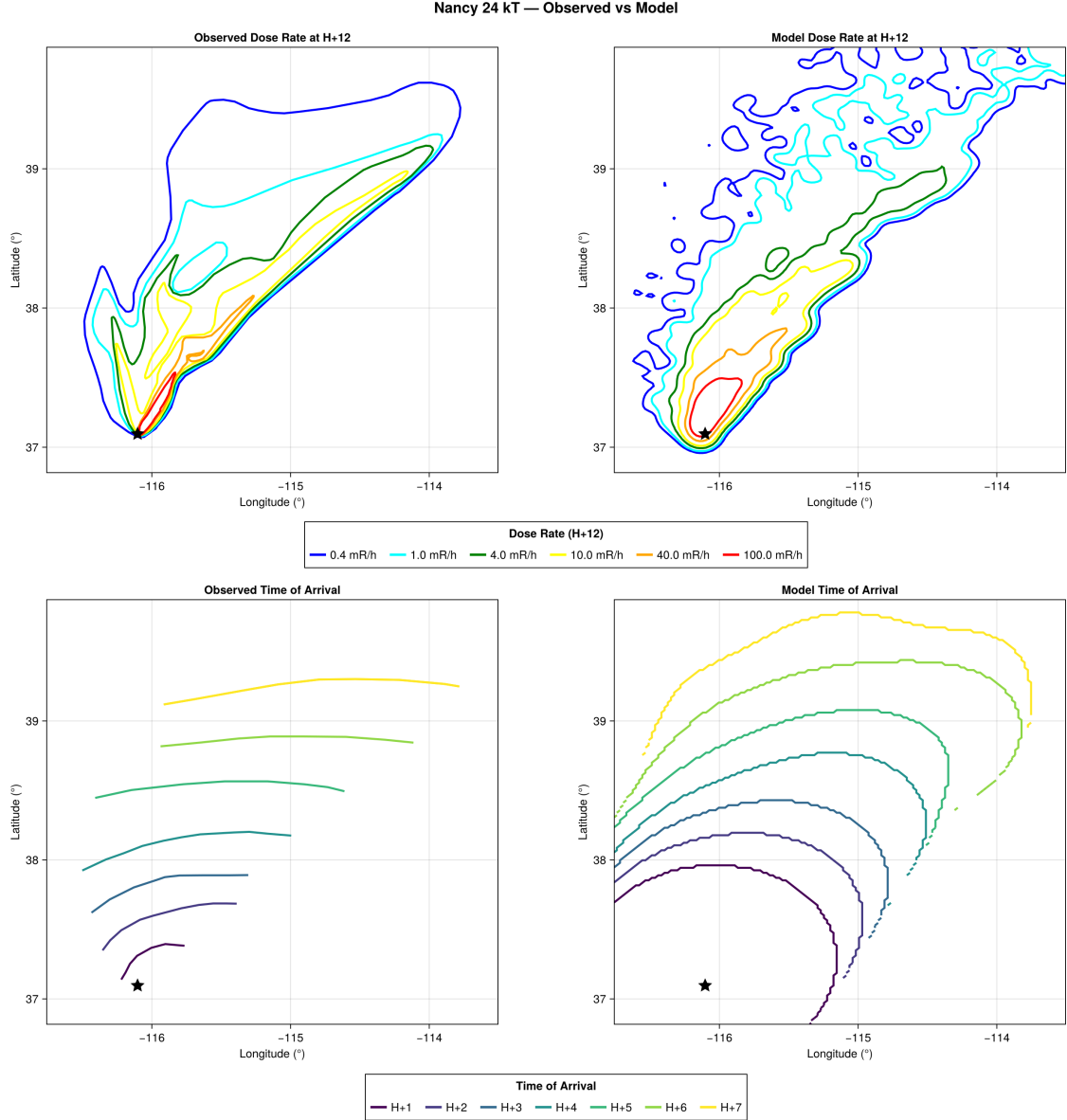


Figure 8: Nancy 24kT: observed (left) vs model (right). Top row: dose rate contours at H+12 for six thresholds (0.4, 1, 4, 10, 40, 100 mR/h). Bottom row: time-of-arrival contours showing plume progression from H+1 through H+7.

5 Smoky 44 kT: Preliminary Results

The same model and OU turbulence scheme were applied to the Plumbbob Smoky test (44kT tower shot, 31 August 1957, Nevada Test Site). Unlike Nancy, where the three-layer boundaries are fixed from NOAA 1984 observations, Smoky’s layer heights were included as tuneable parameters in the CMA-ES optimisation. The best-fit cloud geometry (Figure 9) places the stem top at 1.7km, cap midpoint at 7.4km, and cloud top at 10.4km AGL, with 16.9% of activity in the lower stem, 6.5% in the mid-cap, and 76.6% in the upper cap. The thin grey wireframe in Figure 9 shows the Glasstone & Dolan empirical cloud for comparison; the optimiser infers a broader mid-cap region and a higher cloud top than the empirical formulae predict. ERA5 reanalysis provides the driving meteorology.

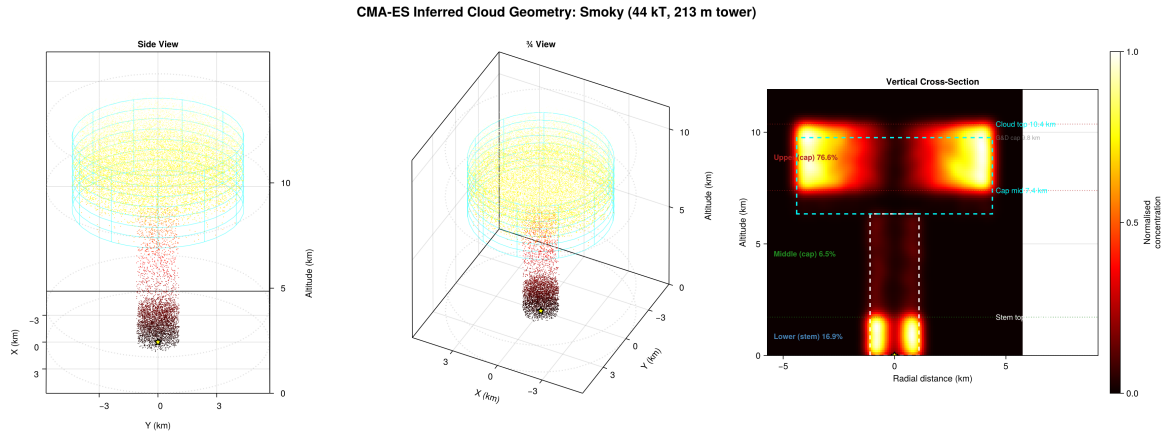


Figure 9: Smoky 44kT CMA-ES-inferred release geometry. Three-layer particle distribution (lower 16.9%, middle 6.5%, upper 76.6%) with coloured wireframes showing the optimised cylinder boundaries. Thin grey outlines show the Glasstone & Dolan empirical cloud for reference.

Table 2 presents the per-threshold FMS scores for Smoky. The current best parameter set achieves a geometric-mean FMS of 44.7%, with the modelled plume correctly reproducing the eastward fallout lobe driven by mid-tropospheric westerlies documented in DNA 6004F. Consortium submissions for Smoky are pending; columns are reserved for comparison once results become available. Calibration is ongoing and we expect further improvement as the parameter space is explored.

Table 2: FMS scores for Smoky 44kT at H+12. Consortium results are pending.

Dose Rate (mR/h)	Our Model	UKMO	BfS	KIT	DSA/MET	DEMA
1	47.4%					
10	44.8%					
50	54.4%					
100	52.5%					
500	40.2%					
1000	32.7%					
Geo. mean	44.7%					

Figure 10 compares observed and modelled fields for both dose rate and time of arrival. The dose rate contours (top row) show the ESE plume axis is well captured, whilst the TOA contours (bottom row) demonstrate that the plume advances at the correct speed, with model H+1 through H+12 contours progressing downwind in agreement with the DASA-1251 [1] survey arcs.

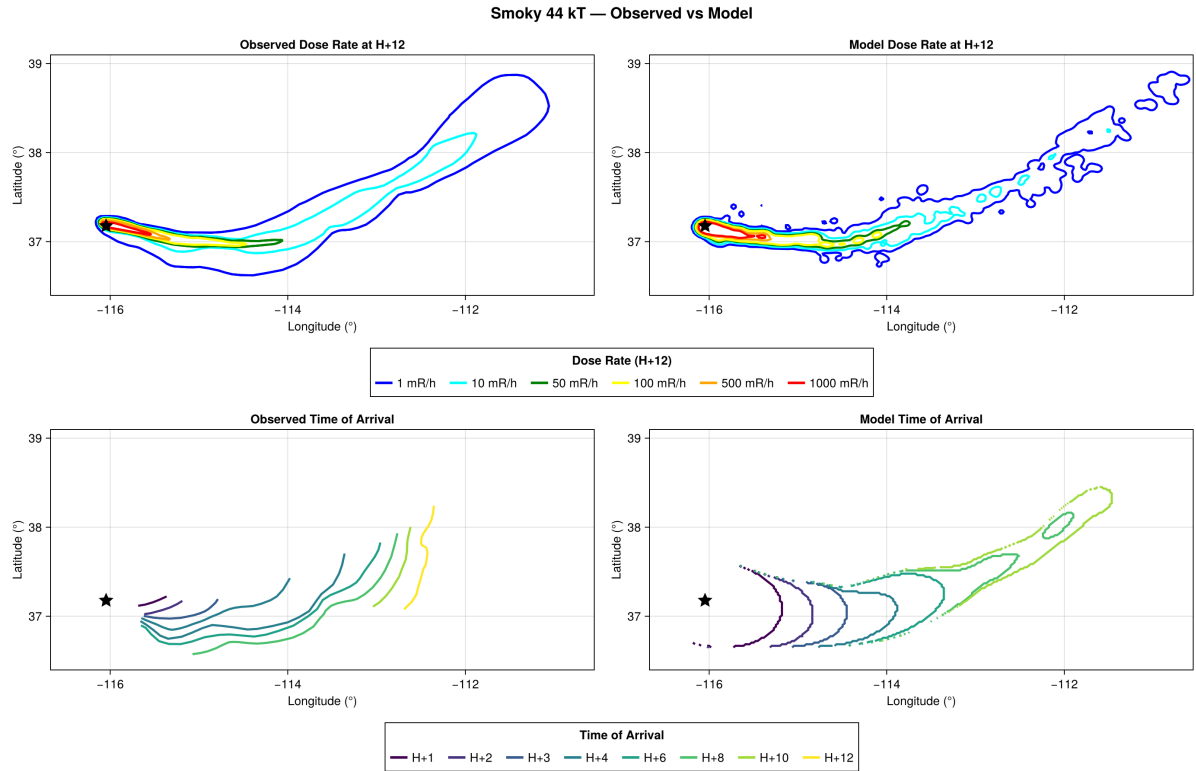


Figure 10: Smoky 44kT: observed (left) vs model (right). Top row: dose rate contours at H+12 for six thresholds (1, 10, 50, 100, 500, 1000 mR/h). Bottom row: time-of-arrival contours showing plume progression from H+1 through H+12.

References

- [1] H. A. Hawthorne, *Compilation of Local Fallout Data from Test Detonations 1945–1962, Extracted from DASA 1251*, Vols. I and II, General Electric Co.–TEMPO, Defense Nuclear Agency, 1979.
- [2] G. E. Uhlenbeck and L. S. Ornstein, “On the Theory of the Brownian Motion,” *Physical Review*, vol. 36, no. 5, pp. 823–841, 1930.
- [3] E. Cunningham, “On the velocity of steady fall of spherical particles through fluid medium,” *Proceedings of the Royal Society A*, vol. 83, no. 563, pp. 357–365, 1910.
- [4] W. Sutherland, “The viscosity of gases and molecular force,” *The London, Edinburgh, and Dublin Philosophical Magazine and Journal of Science*, vol. 36, no. 223, pp. 507–531, 1893.
- [5] L. Zhang, S. Gong, J. Padro, and L. Barrie, “A size-segregated particle dry deposition scheme for an atmospheric aerosol module,” *Atmospheric Environment*, vol. 35, no. 3, pp. 549–560, 2001.